
AWS VPC WITH VARIABLE-SIZE SUBNETS

Objective

Design and deploy a **highly complex AWS VPC architecture** using Terraform, where each subnet has a different size based on the required IP capacity. This setup simulates a **production-grade environment** supporting both public-facing and private resources.

Task Requirements

1. Create a New VPC

- Name the VPC "**asymmetric-vpc**".
- Use a **/16 CIDR block**.
- Enable **DNS resolution** and **DNS hostnames**.
- Tag all resources with:

Environment = "production"

Owner = "network-team"

Project = "asymmetric-vpc-build"

CostCenter = "AWS-Networking"

2. Create SIX Subnets with Different Capacities

Design six subnets with **varying IP capacities** representing real enterprise workloads. Each subnet should have a unique **CIDR size** according to expected IP needs.

Subnet	Required IP Capacity	CIDR Size
Public Subnet A	~256 IPs	/24
Public Subnet B	~4,096 IPs	/20
Public Subnet C	~8,192 IPs	/19
Private Subnet A	~1,024 IPs	/22
Private Subnet B	~512 IPs	/23
Private Subnet C	~8,192 IPs	/19

Requirements for Subnets

- **Public Subnets**
 - Deploy in **different Availability Zones (AZs)**.
 - Automatically assign public IPs on launch.
 - Intended for **internet-facing resources** (e.g., load balancers, bastion hosts).

- **Private Subnets**

- Deploy in **different AZs**.
- No automatic public IP assignment.
- Intended for **internal resources** (e.g., databases, application servers).
- Internet access via NAT Gateway(s).

Focus on **IP capacity planning** and **AZ distribution** rather than specifying exact CIDR blocks.

3. Create an Internet Gateway

- Name the IGW appropriately (e.g., "asym-igw").
 - Attach it to the VPC.
 - Ensure public subnets can communicate directly with the internet.
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4. Create NAT Gateway(s)

- Choose one of the following strategies:
 - **Single NAT Gateway** in the largest public subnet (simple).
 - **Multiple NAT Gateways**, one per AZ (recommended for high availability).

- Each NAT Gateway must have its own Elastic IP.
 - Private subnets must route traffic through the NAT Gateway for outbound internet connectivity.
 - Document and justify your chosen design.
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5. Create Route Tables

Public Route Tables

- One route table per public subnet.
- Default route **0.0.0.0/0** → Internet Gateway.
- Associate **only** with its corresponding public subnet.

Private Route Tables

- One route table per private subnet.
- Default route → NAT Gateway serving that AZ.
- Associate **only** with its corresponding private subnet.

No sharing of route tables between subnets.

6. Subnet-to-Route Table Associations

- Each subnet has a **dedicated route table**.
 - Public subnets → public route tables.
 - Private subnets → private route tables.
 - Ensure **clear isolation** between public and private subnets.
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7. Tagging Requirements

- All resources must have consistent tags:

Environment = "production"

Owner = "network-team"

Project = "asymmetric-vpc-build"

CostCenter = "AWS-Networking"

- Add Tier = "public" or Tier = "private" for clarity.
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